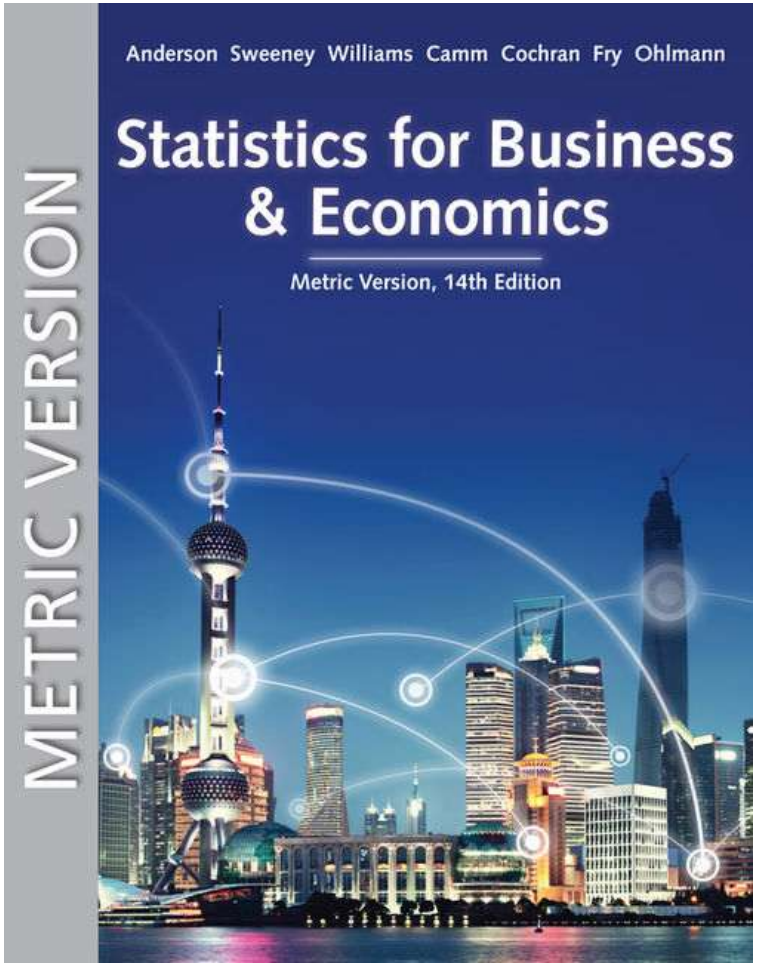
Statistics for Business and Economics (14e) Metric Version

Anderson, Sweeney, Williams, Camm, Cochran, Fry, Ohlmann

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Chapter 1 - Data and Statistics

1.1 - Applications in Business and Economics

1.2 - Data

1.3 - Data Sources

1.4 - Descriptive Statistics

1.5 - Statistical Inference

1.6 - Analytics

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What Is Statistics?

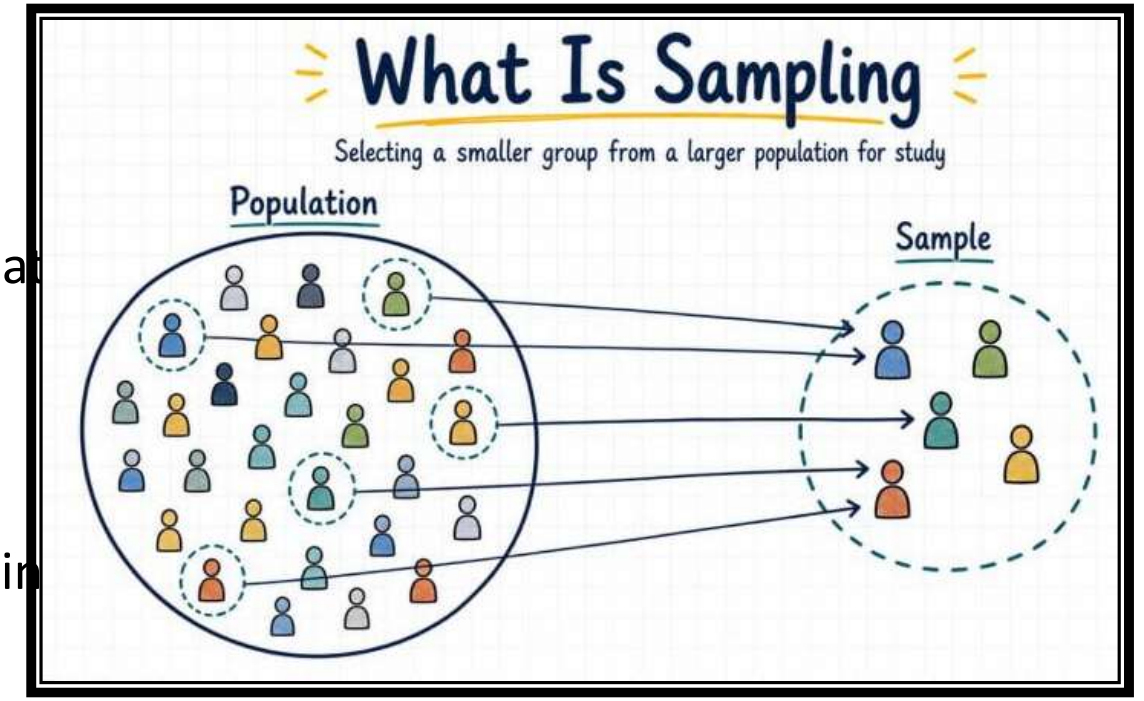
- The term statistics can refer to ***numerical facts*** (統計數據)
  - Such as averages, medians, percentages, and maximums that help us understand a variety of business and economic situations.
- Statistics can also refer to the ***art and science***
  - Including collecting, analyzing, presenting, and interpreting data

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Applications in Business and Economics (1 of 2)

Accounting

- Public accounting firms use statistical sampling procedures when conducting audits for their clients.



Economics

- Economists use statistical informat
economy or some aspect of it.

Finance

- Financial advisors use price-earnin
investment advice.

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Applications in Business and Economics (1 of 2)

Accounting

- Public accounting firms use statistical sampling procedures when conducting audits for their clients.

Economics

- Economists use statistical information in making forecasts about the future of the economy or some aspect of it.

Finance

- Financial advisors use price-earnings ratios and dividend yields to guide their investment advice.

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Applications in Business and Economics (2 of 2)

Marketing

- Electronic point-of-sale scanners at retail checkout counters are used to collect data for a variety of marketing research applications.

Production

- A variety of statistical quality control charts are used to monitor the output of a production process.

Information Systems

- A variety of statistical information helps administrators assess the performance of computer networks.

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Data and Data Sets

- **Data** are the facts and figures collected, analyzed, and summarized for presentation and interpretation.
- All the data collected in a particular study are referred to as the **data set** for the study.



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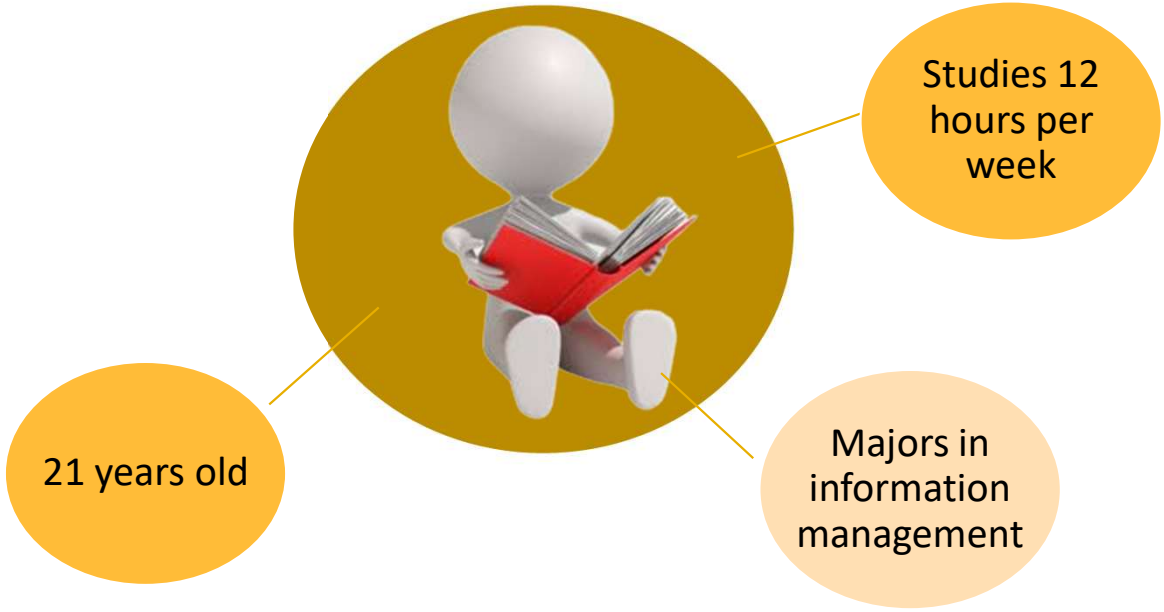
Elements, Variables, and Observations

- Elements are the entities on which data are collected.
- A variable is a characteristic of interest for the elements.
- The set of measurements obtained for a particular element is called an observation.

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Elements, Variables, and Observations

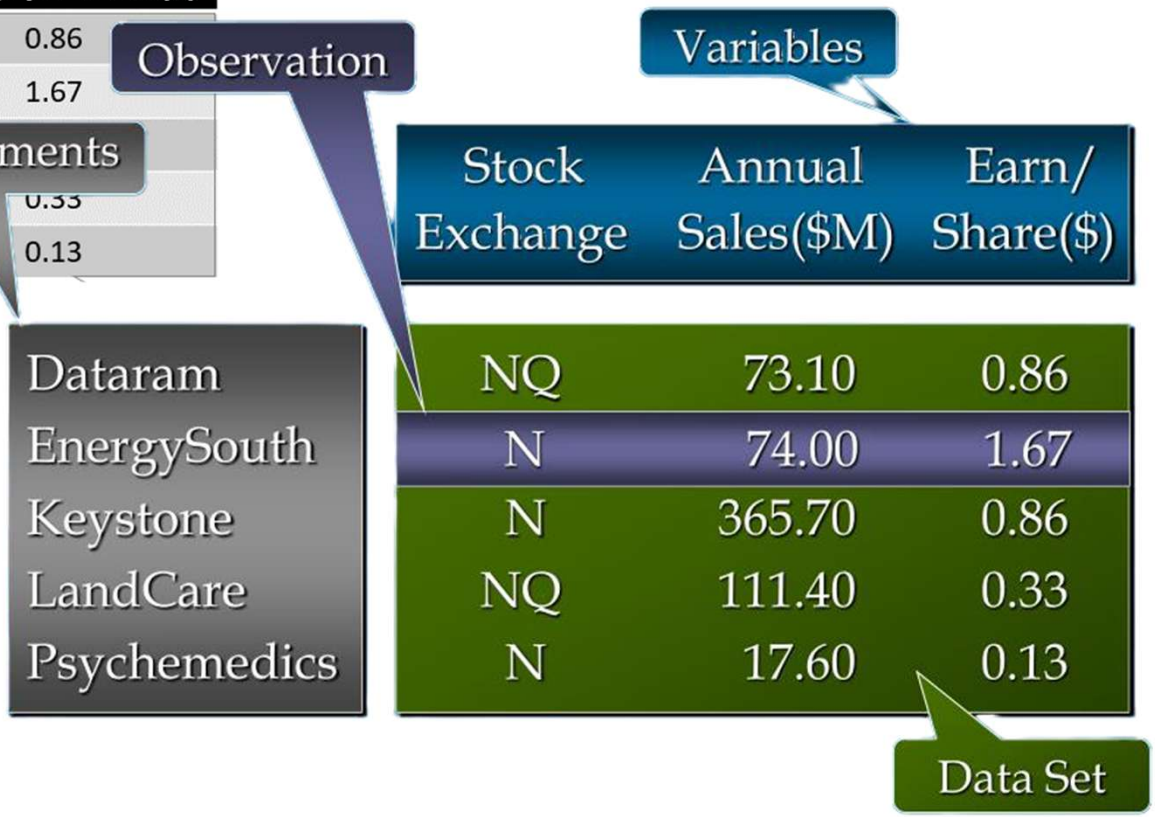
- A data set with *n* elements contains *n* observations.
- The total number of data values in a complete data set is the number of elements multiplied by the number of variables.



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Data, Data Sets, Elements, Variables, and Observations

Company	Stock Exchange	Annual Sales ($M)	Earnings per share ($)
Dataram	NQ	73.10	0.86
EnergySouth	N	74.00	1.67
Keystone	N	365.70	0.86
LandCare	NQ	111.40	0.33
Psychemedics	N	17.60	0.13



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Scales of Measurement

- Scales of measurement include
  - **Nominal**
  - **Ordinal**
  - **Interval**
  - **Ratio**
- The scale determines the amount of information contained in the data.
- The scale indicates the data summarization that are most appropriate.
- The scale indicates the statistical analyses that are most appropriate.

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Scales of Measurement

Nominal scale

- Data are labels or names used to identify an attribute of the element.
- A nonnumeric label or numeric code may be used.

Example

Students of a university are classified by the college in which they are enrolled using a nonnumeric label such as College of Management, College of Engineering, College of Medicine, and so on.

1 = Management
2 = Engineering
3 = Medicine

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Scales of Measurement

Ordinal scale

- The data have the properties of nominal data and the order or rank of the data is meaningful.
- A nonnumeric label or numeric code may be used.

Example



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Scales of Measurement



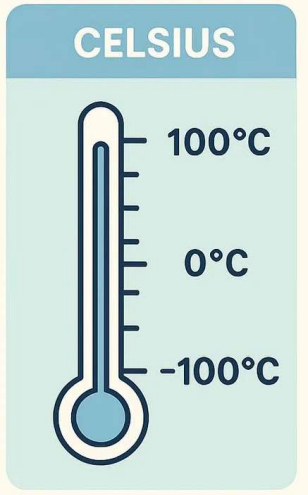
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Scales of Measurement

Interval scale

- The data have the properties of ordinal data, and the interval between observations is expressed in terms of a fixed unit of measure.
- Interval data are always numeric.

Example: Temperature measured in degrees Celsius



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Scales of Measurement

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	
Ranks categories in order		✓	✓	
Has known, equal intervals			✓	

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Scales of Measurement

Ratio scale

- Data have all the properties of interval data and the ratio of two values is meaningful.
- Ratio data are always numerical.
- Zero value is included in the scale.

Example:

Price of a book at a retail store is $200, while the price of the same book sold online is $100. The ratio property shows that retail stores charge twice the online price.

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Scales of Measurement

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

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Categorical and Quantitative Data

- Data can be further classified as being categorical or quantitative.
- The statistical analysis that is appropriate depends on whether the data for the variable are categorical or quantitative.
- In general, there are more alternatives for statistical analysis when the data are quantitative.

Categorical data → Which group or category?
Quantitative data → How much or how many?

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Categorical Data

- Labels or names are used to identify an attribute of each element
- Often referred to as qualitative data
- Use either the nominal or ordinal scale of measurement
- Can be either numeric or nonnumeric
- Appropriate statistical analyses are rather limited

NOMINAL

Labels only, no order

Fruits:



Apple Banana Orange

ORDINAL

Ordered categories, no equal intervals

Ratings:

Poor

Fair

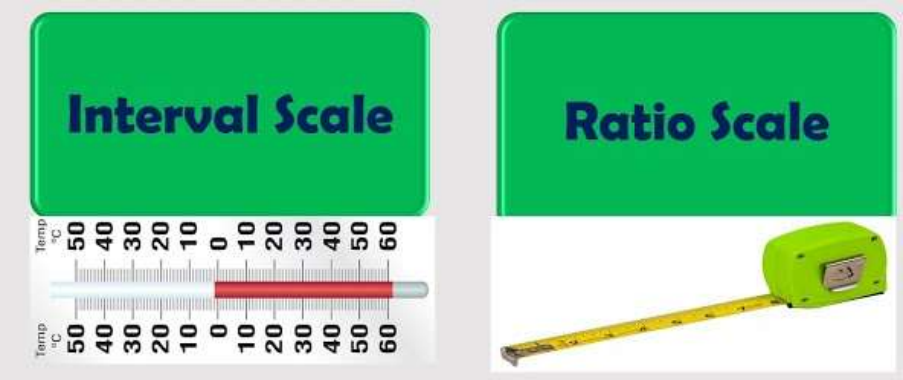
Good

Excellent

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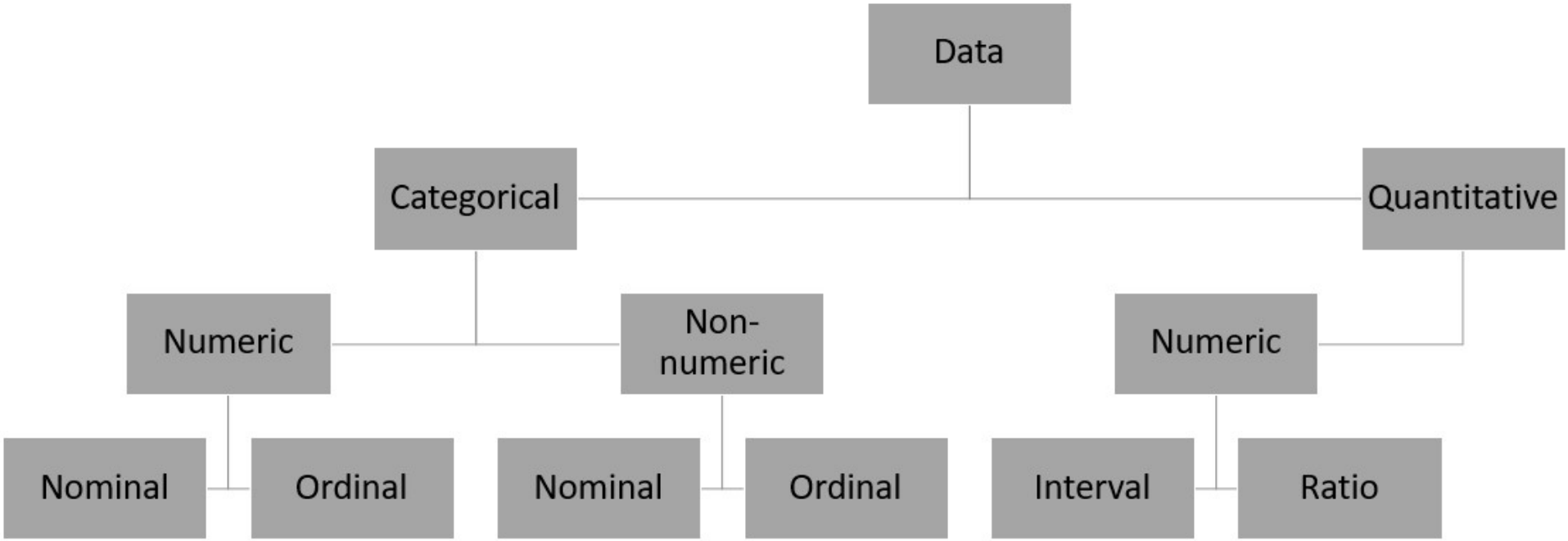
Quantitative Data

- Quantitative data indicate how many or how much.
- Quantitative data are always numeric.
- Ordinary arithmetic operations are meaningful for quantitative data.
- Use either the interval or the ratio scale of measurement



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Scales of Measurement



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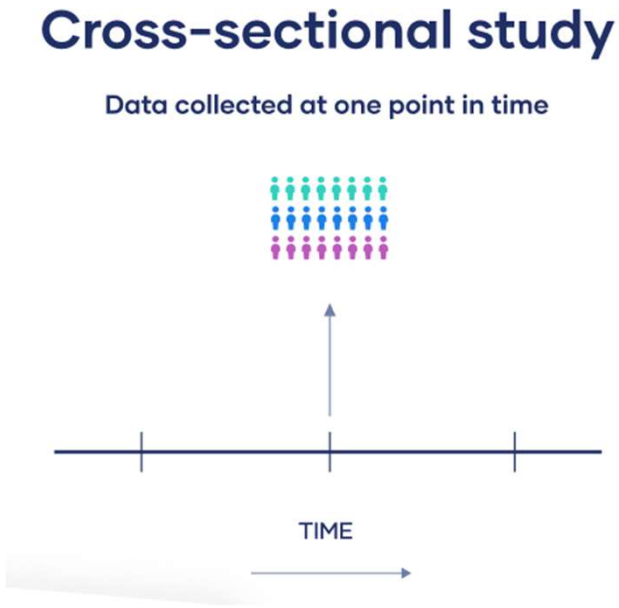
Cross-Sectional Data

Cross-sectional data is collected at the same or approximately the same point in time.

Example

Data detailing the number of building permits issued

in September 2025 in each city across Taiwan



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Time Series Data (1 of 2)

Time series data is collected over several time periods.

Example

Data detailing the number of building permits issued in Taipei city in each of the last 36 months.



Graphs of time series data help analysts understand

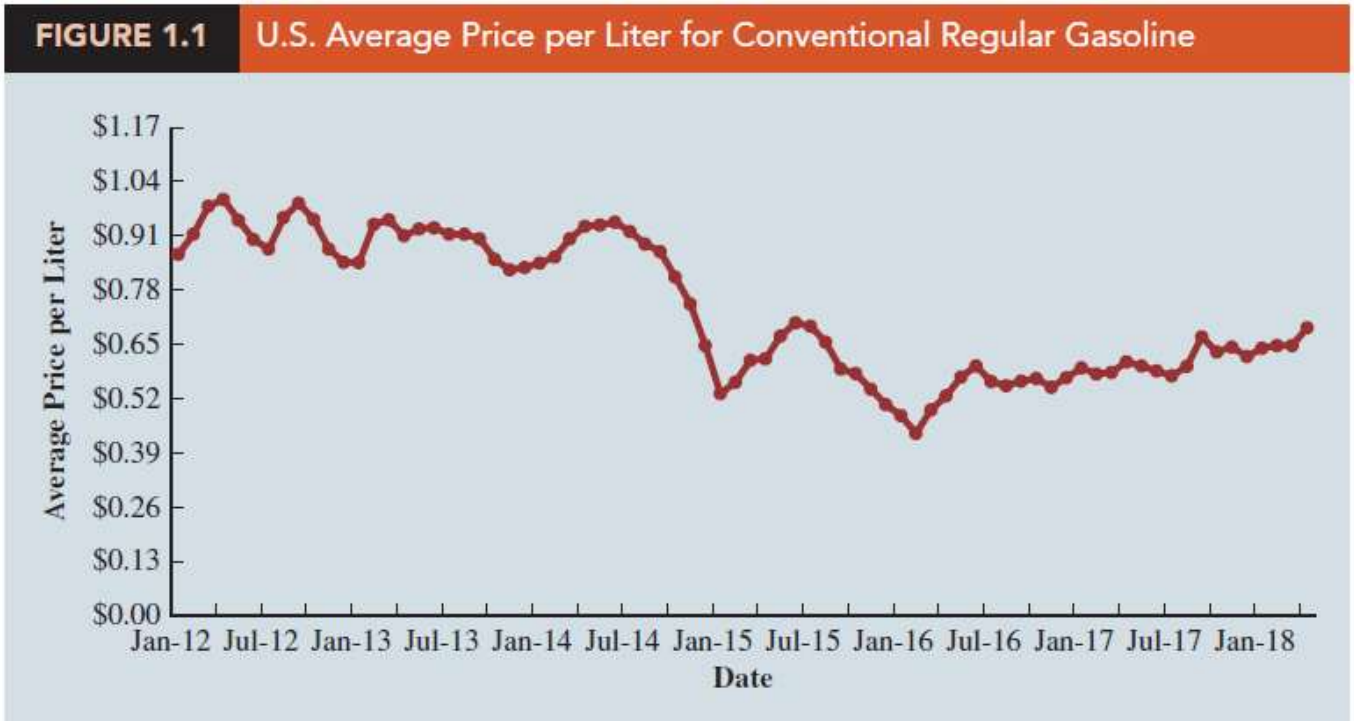
- what happened in the past
- identify any trends over time, and
- project future levels for the time series

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Time Series Data (2 of 2)

Graph of Time Series Data

FIGURE 1.1 U.S. Average Price per Liter for Conventional Regular Gasoline



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Data Sources (1 of 4)

Existing Sources

- Internal company records – almost any department
- Business database services – Dow Jones & Co.
- Government agencies – U.S. Department of Labor
- Industry associations – Travel Industry Association of America
- Special-interest organizations – Graduate Management Admission Council (GMAT)
- Internet – more and more firms

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Data Sources (2 of 4)

Data Available From Internal Company Records

Record	Some of the Data Available
Employee records	Name, address, social security number
Production records	Part number, quantity produced, direct labor cost, material cost
Inventory records	Part number, quantity in stock, reorder level, economic order quantity
Sales records	Product number, sales volume, sales volume by region
Credit records	Customer name, credit limit, accounts receivable balance
Customer profile	Age, gender, income, household size

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Data Sources (3 of 4)

Statistical Studies – Observational

- In observational (nonexperimental) studies no attempt is made to control or influence the variables of interest.
- Example - Survey
- Studies of smokers and nonsmokers are observational studies because researchers do not determine or control who will smoke and who will not smoke.

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Data Sources (4 of 4)

Statistical Studies – Experimental

- In experimental studies the variable of interest is first identified. Then one or more other variables are identified and controlled so that data can be obtained about how they influence the variable of interest.
- Example - Randomized controlled trial (RCT) in clinical research

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Data Acquisition Considerations

Time Requirement

- Searching for information can be time consuming.
- Information may no longer be useful by the time it is available.

Cost of Acquisition

- Organizations often charge for information even when it is not their primary business activity.

Data Errors

- Using any data that happen to be available or were acquired with little care can lead to misleading information.

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Descriptive Statistics

- Most of the statistical information in newspapers, magazines, company reports, and other publications consists of data that are summarized and presented in a form that is easy to understand.
- Such summaries of data, which may be tabular, graphical, or numerical, are referred to as descriptive statistics.

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Example: Hudson Auto Repair

- The manager of Hudson Auto would like to have a better understanding of the cost of parts used in the engine tune-ups performed in her shop. She examines 50 customer invoices for tune-ups.

- Sample of Parts Cost ($) for 50 Tune-ups

91	78	93	57	75	52	99	80	97	62
71	69	72	89	66	75	79	75	72	76
104	74	62	68	97	105	77	65	80	109
85	97	88	68	83	68	71	69	67	74
62	82	98	101	79	105	79	69	62	73

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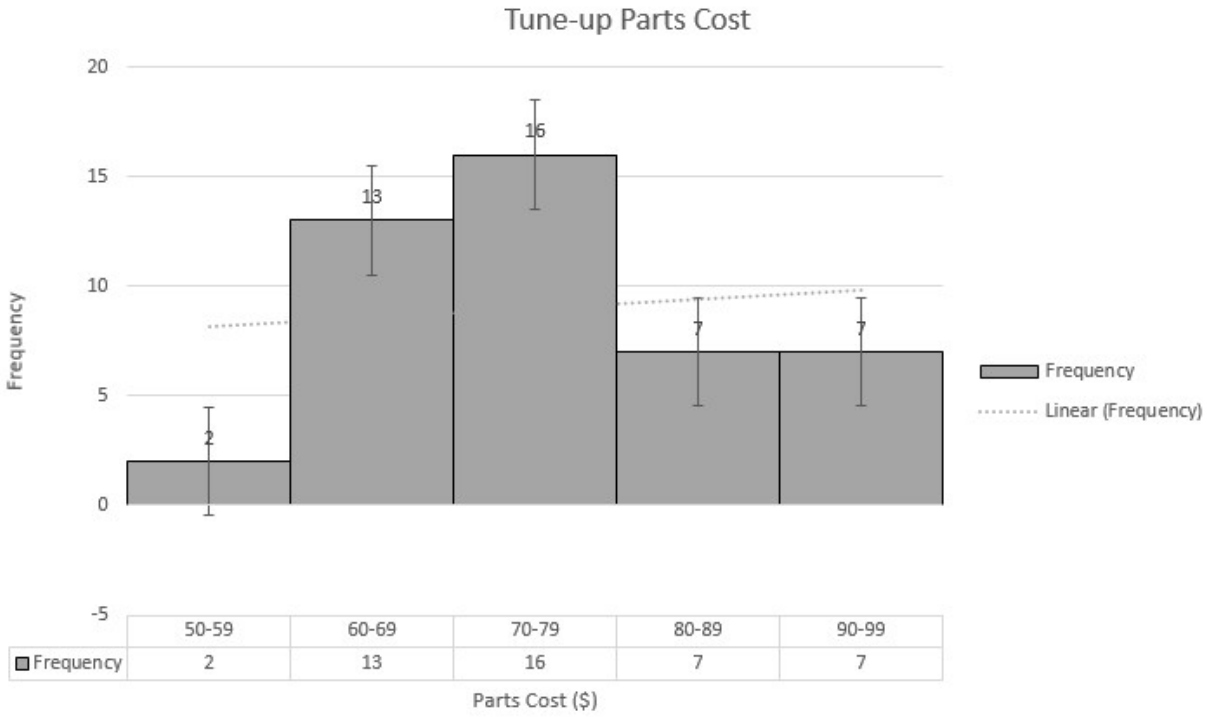
Tabular Summary: Frequency and Percent Frequency

Parts Cost ($)	Frequency	Percent Frequency
50-59	2	4%
60-69	13	26%
70-79	16	32%
80-89	7	14%
90-99	7	14%
100-109	5	10%
TOTAL	**50**	**100%**

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Graphical Summary: Histogram

Example: Hudson Auto



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Numerical Descriptive Statistics

- The most common numerical descriptive statistic is the mean (or average).
- The mean demonstrates a measure of the central tendency, or central location of the data for a variable.
- Hudson's mean cost of parts, based on the 50 tune-ups studied is $79 (found by summing up the 50 cost values and then dividing by 50).

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Statistical Inference

Population: The set of all elements of interest in a particular study.

Sample: A subset of the population.

Statistical inference: The process of using data obtained from a sample to make estimates and test hypotheses about the characteristics of a population.

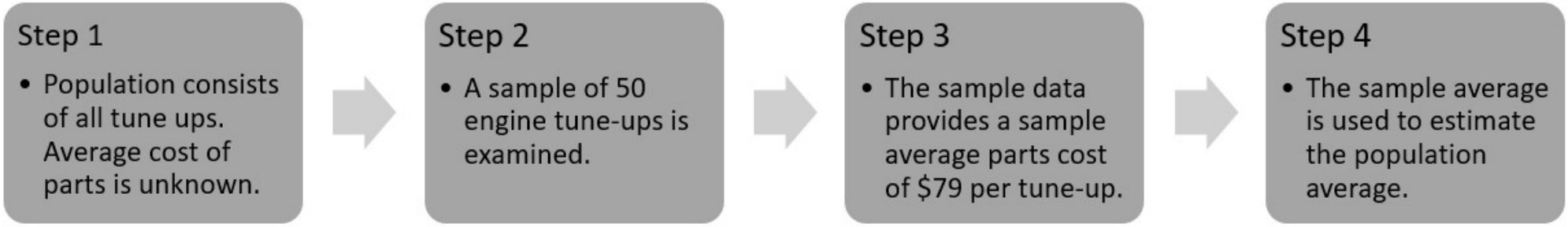
Census: Collecting data for the entire population.

Sample survey: Collecting data for a sample.

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Process of Statistical Inference

Example: Hudson Auto



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Analytics

Analytics is the scientific process of transforming data into insight for making better decisions.

Techniques:

- Descriptive analytics: This describes what has happened in the past.

- Predictive analytics: Use models constructed from past data to predict the future or to assess the impact of one variable on another.

- Prescriptive analytics: The set of analytical techniques that yield a best course of action.

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